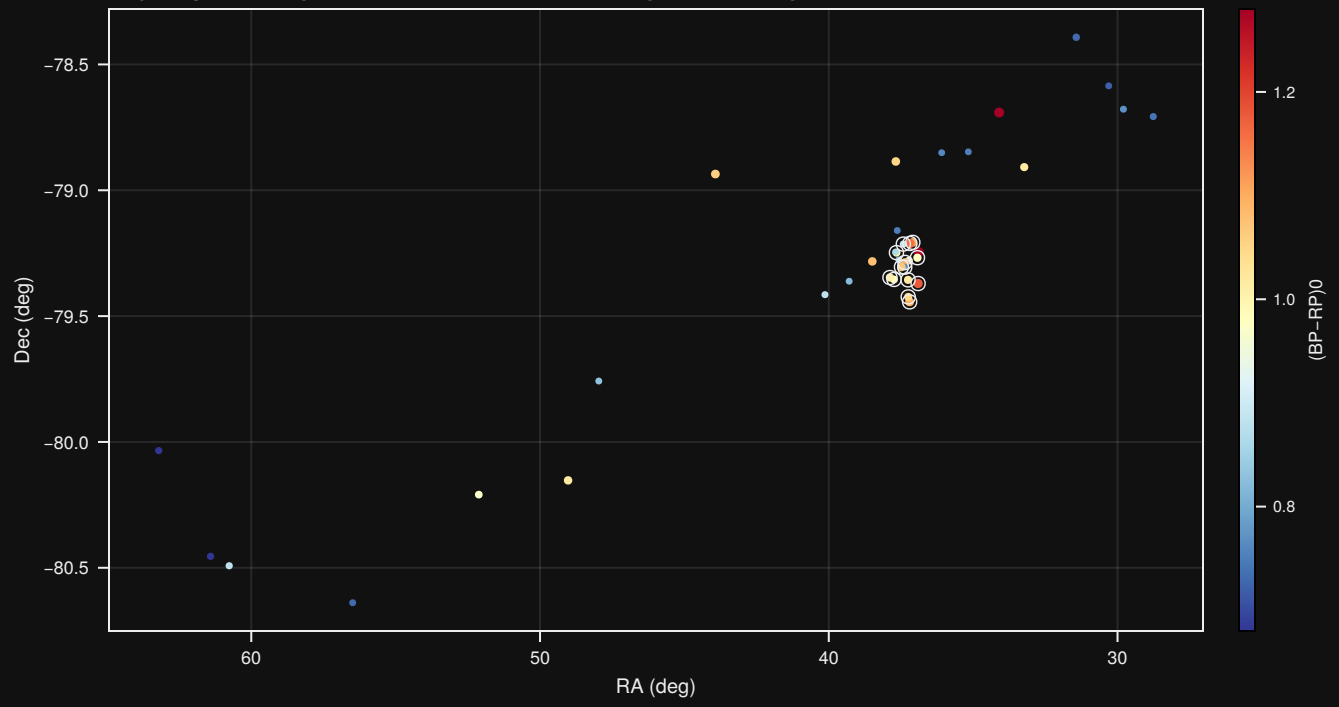
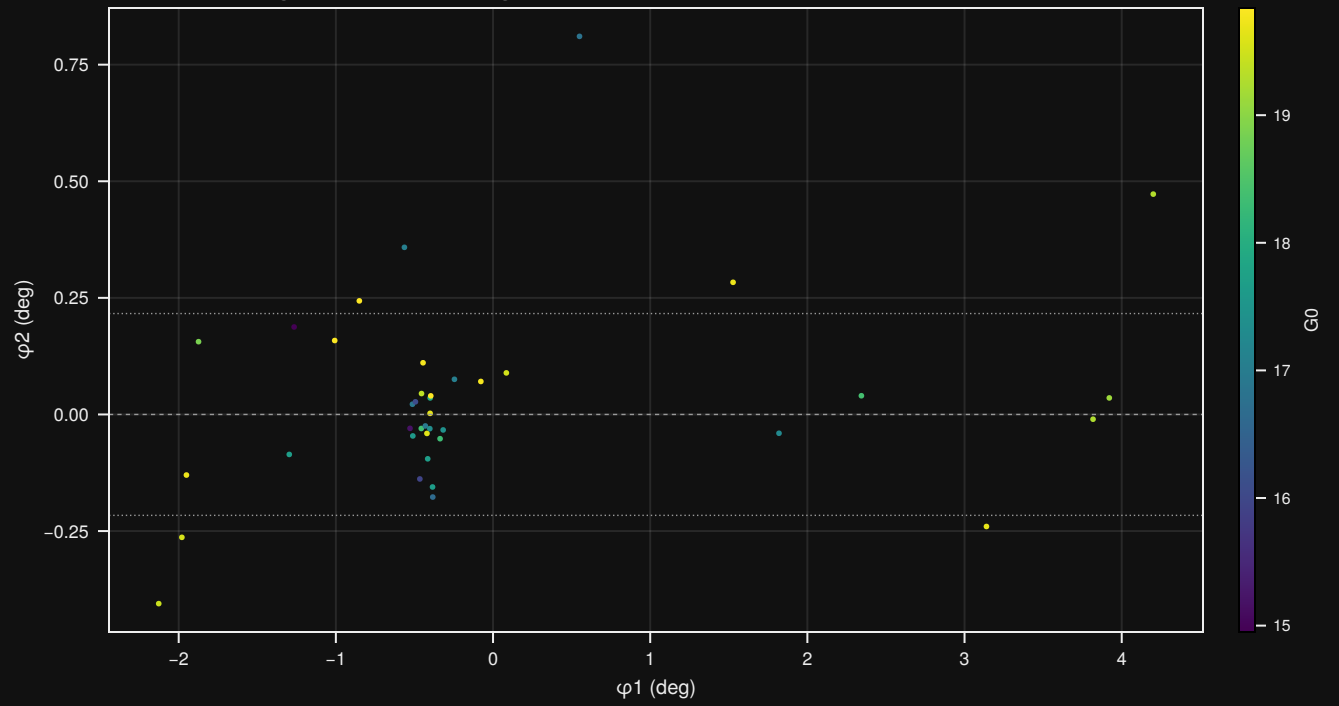
Hydrus

MEMBERSHIP	
Stars in catalog	39
with Gaia GSP-Phot	18 (46%)
with radial velocity	14
Stream id (sID)	11
GEOMETRY · great-circle fit	
Centroid RA, Dec	39.9, -79.3 deg
Centroid l, b	297.1, -36.8 deg
Pole RA, Dec	65.6, 9.5 deg
Length (φ1, 2–98%)	6.3 deg
Width σ(φ2), robust	0.11 deg
KINEMATICS	
pmRA* med ± σ	3.83 ± 0.23 mas/yr
pmDE med ± σ	-1.49 ± 0.23 mas/yr
μφ1* med ± σ	4.07 ± 0.20 mas/yr
μφ2 med ± σ	0.44 ± 0.13 mas/yr
Parallax (cat.) med	0.034 ± 0.136 mas
Parallax (wtd, ZP-corr.)	0.040 ± 0.016 mas
Distance	d > 14.0 kpc
Physical length	—
RV median ± σ(MAD)	80.8 ± 2.4 km/s
RV gradient dV/dφ1	24.07 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	14.9–19.8 (18.4)
(BP−RP)0 median	0.97
GSP-Phot [M/H] (n=18)	-1.71 ± 1.72
GSP-Phot Teff median	5177 K
RUWE > 1.4	5.6 %
PM clusters (largest)	none
IBATA+2024 · TABLE 3	
n catalog | paper	39 | 39
n_RV catalog | paper	14 | 14
<[Fe/H]> paper	-3.48
Stellar mass (paper)	7700 Msun
Pole offset vs paper	1.7 deg
Remark	New stream of known structure

Sky map · catalog members (n = 39) · size ∝ brightness · ring = has RV

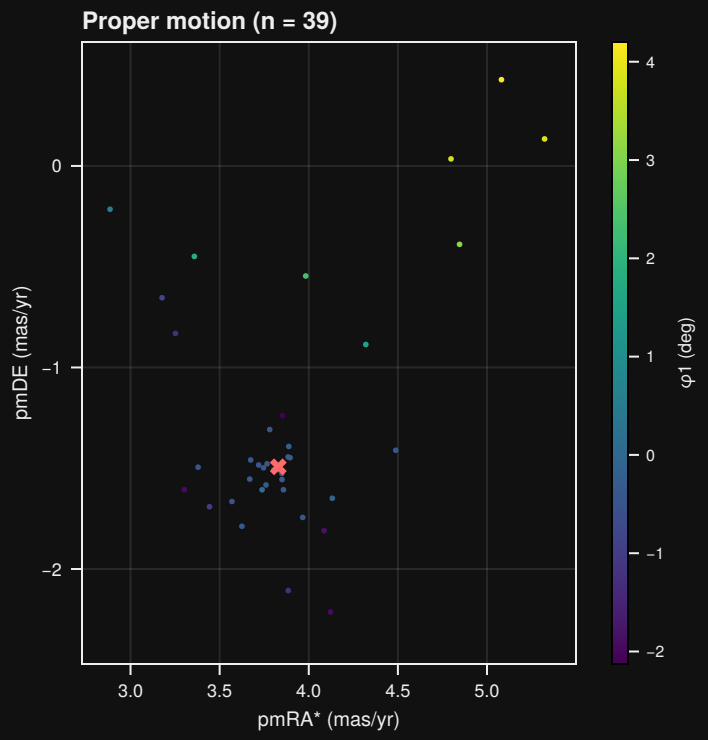


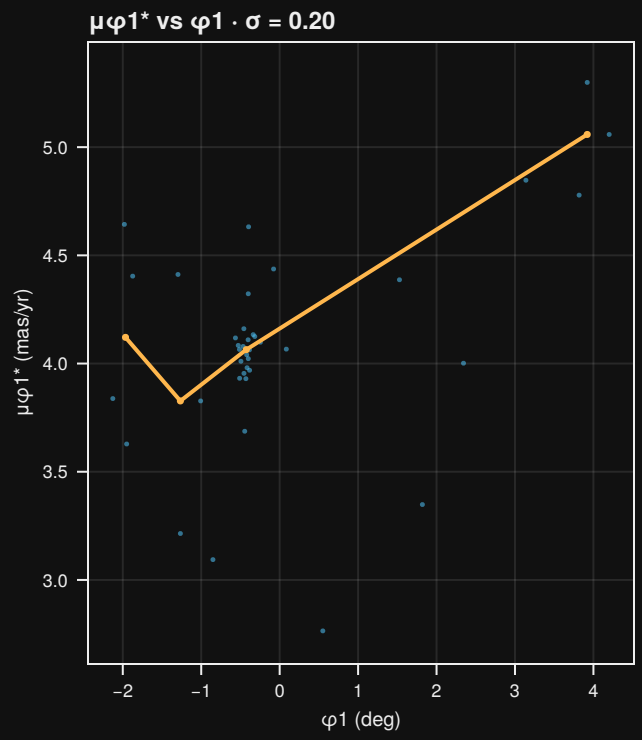
Stream frame (great-circle fit) · length 6.3° · width σ 0.11°

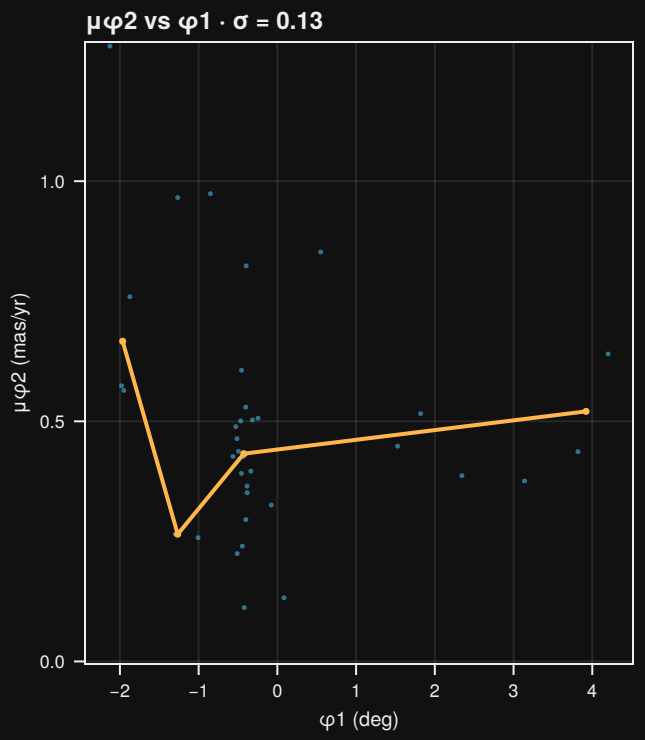


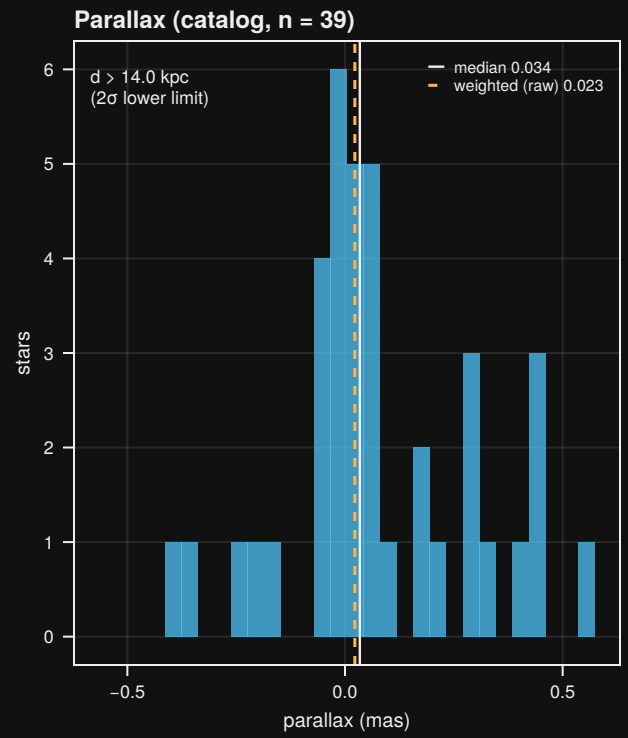
Hydrus

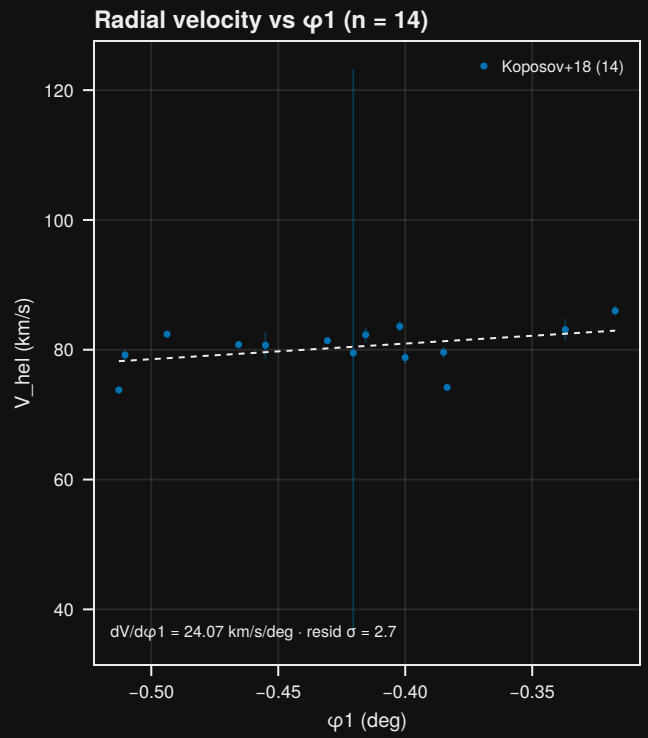
Kinematics · proper motion · parallax · radial velocity

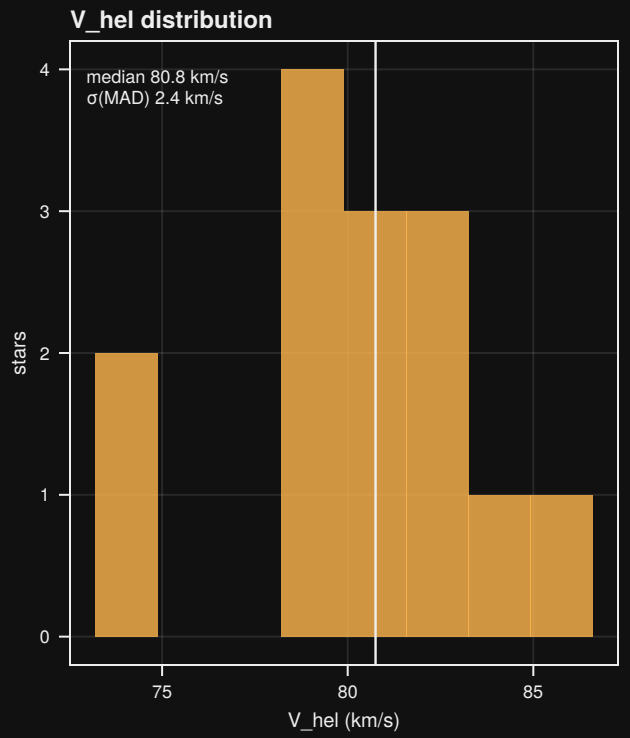






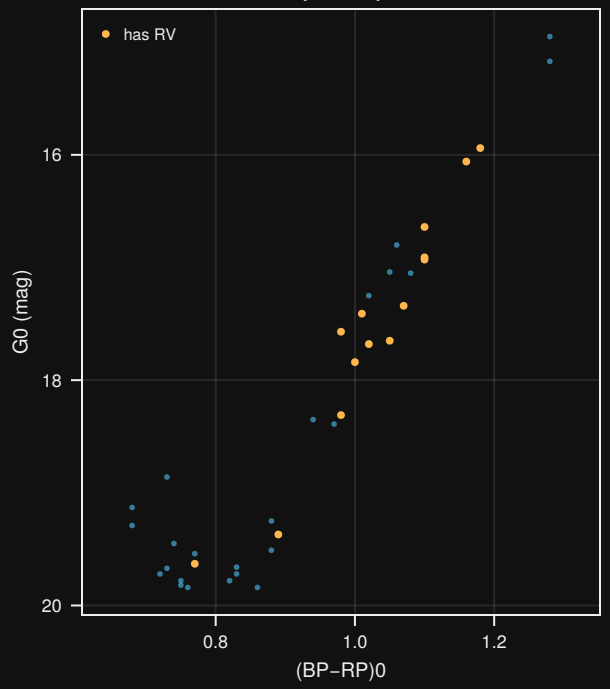




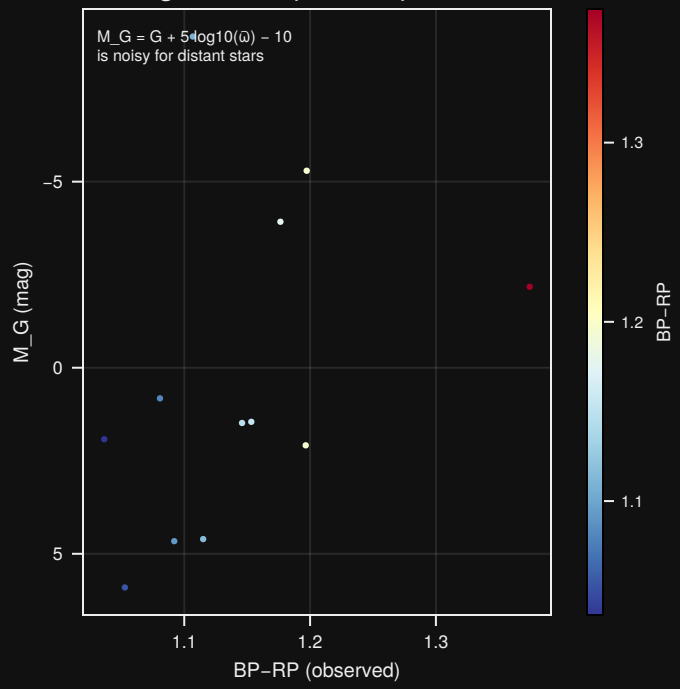


Hydrus

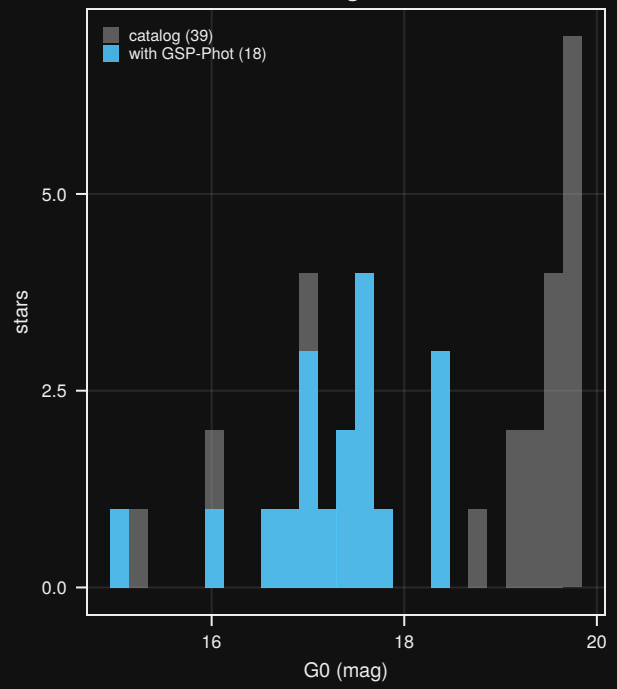
CMD · dereddened (n = 39)



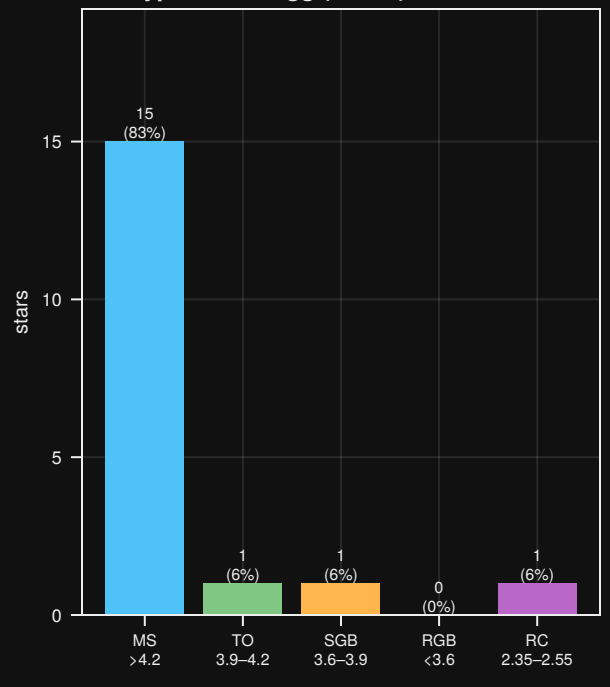
HR diagram · ϖ>0 (n = 12/18)



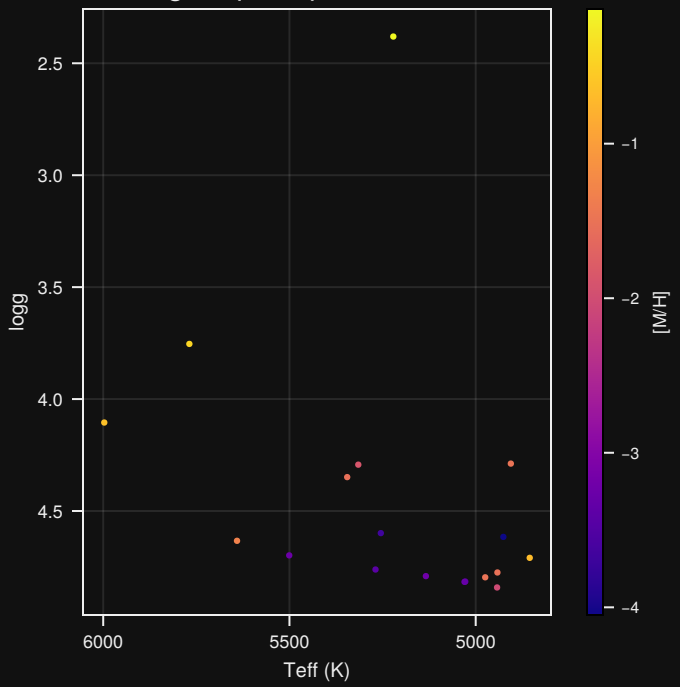
G0 & GSP-Phot coverage



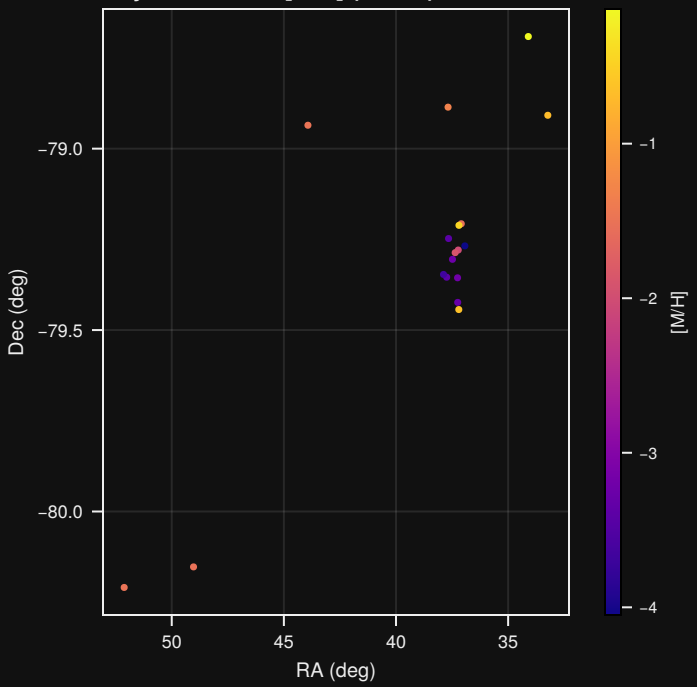
Star types from logg (n = 18)



Kiel diagram (n = 18)



Sky · GSP-Phot [M/H] (n = 18)

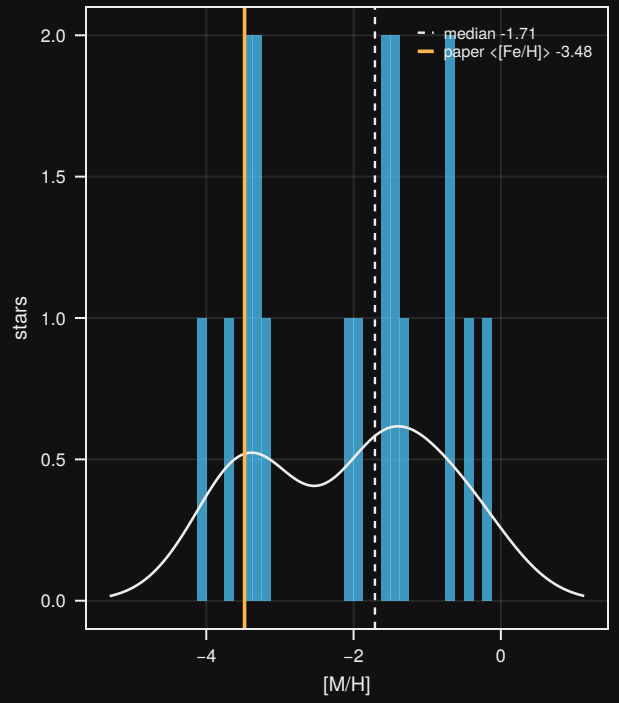


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

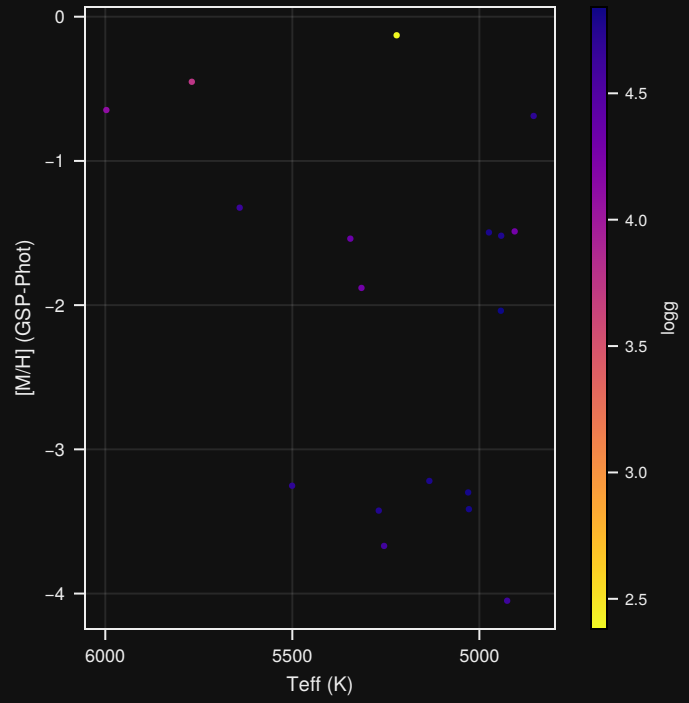
Hydrus

Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 18)



Teff vs [M/H] (n = 18)



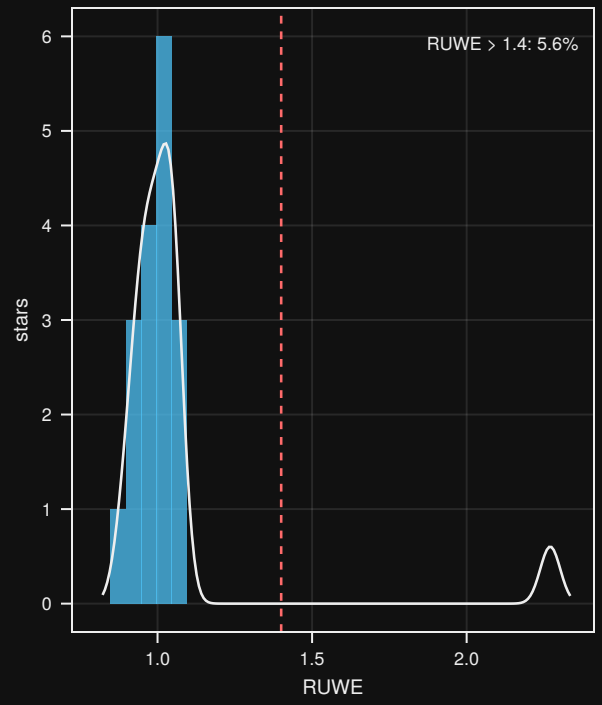
FLAME age (n = 1)

FLAME ages for only 1 star(s)
(FLAME needs good parallaxes)

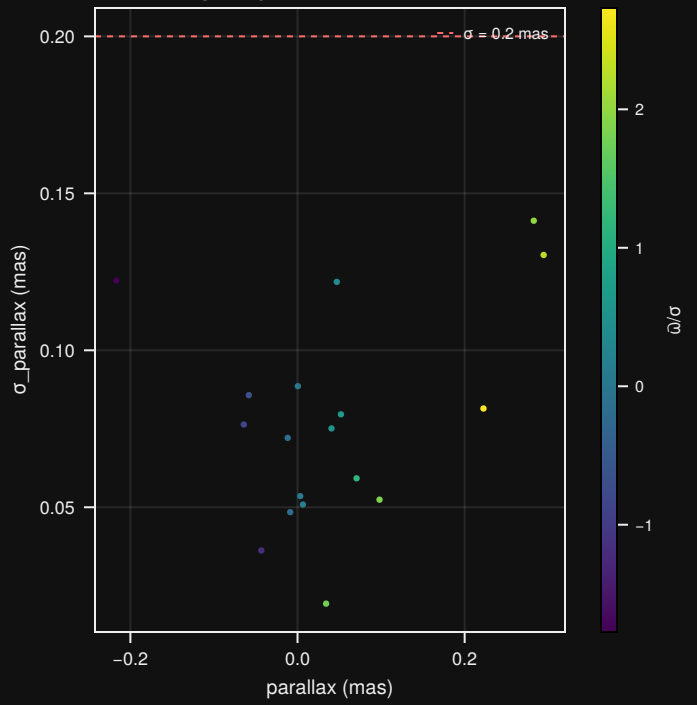
Age–metallicity (n = 1)

n < 3

RUWE (n = 18)

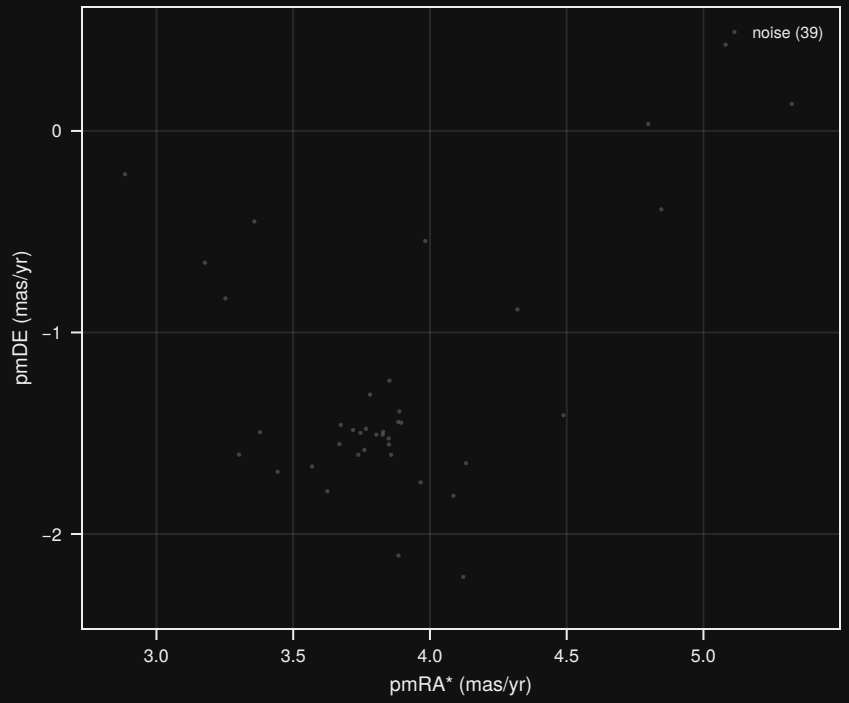


Parallax quality (n = 18)

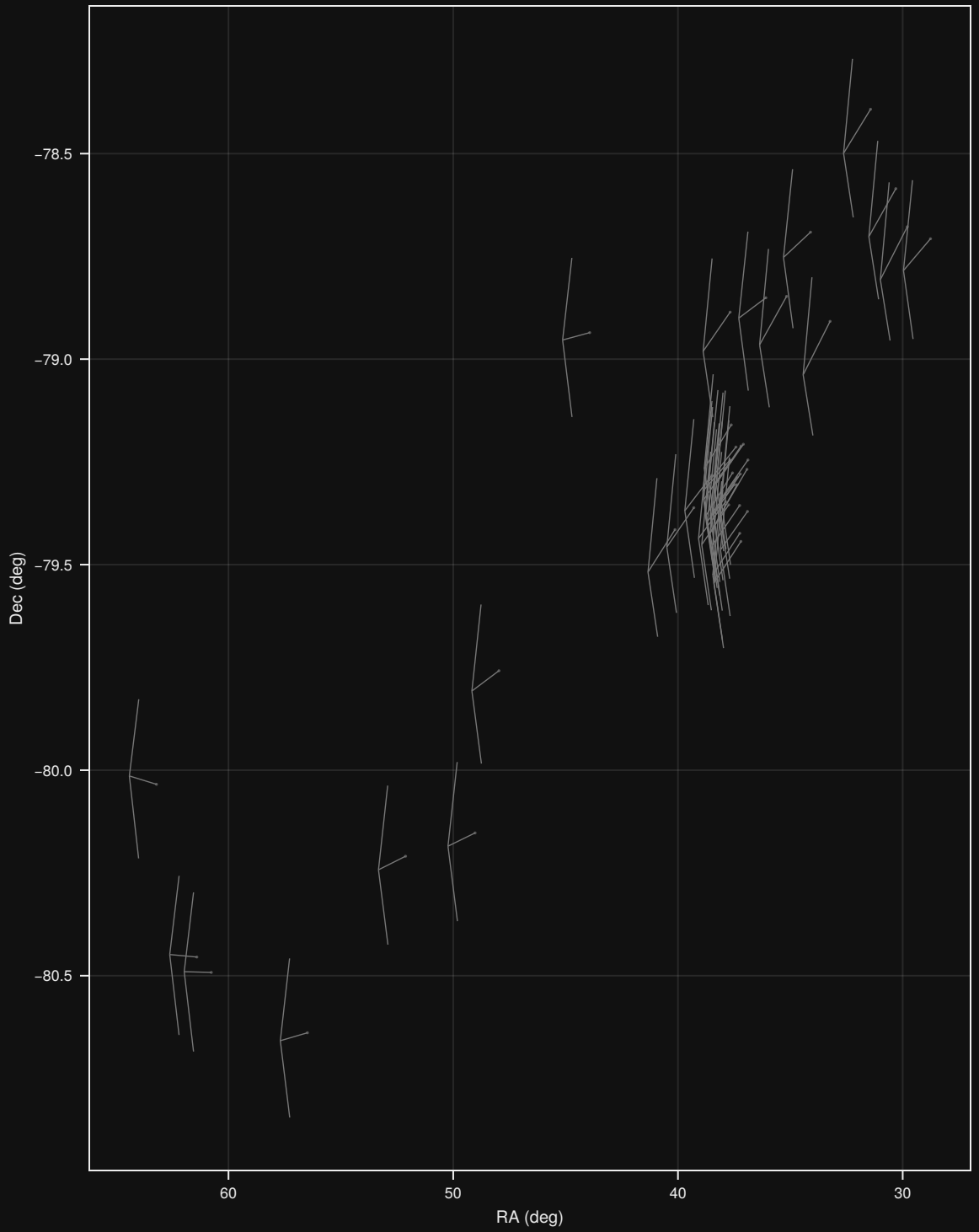


Hydrus

PM clustering · HDBSCAN · 0 cluster(s) · noise 100%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	39	3.83	-1.49	0.23	-1.71	14